

GW1NS & GW1NSR & GW1NSER series of FPGA Products Schematic Manual

Introduction

You should follow some rules for circuit board design when using GW1NS & GW1NSR & GW1NSER series of FPGA products. This manual describes the characteristics of GW1NS & GW1NSR & GW1NSER series of FPGA products. The main contents of this manual are as follows:

- Power Supply
- JTAG Download
- MSPI
- Clock Pin
- Differential Pin
- RECONFIG_N, READY, DONE
- MODE
- JTAGSEL_N
- FASTRD_N
- Dual-purpose Pin
- FPGA External Crystal Oscillator Circuit Reference
- Bank Voltage
- Configuration Modes Supported by Each Device
- MIPI
- Pin Distribution

Power Supply

Overview

GW1NS & GW1NSR & GW1NSER series of FPGA products support LV version. The voltage includes core voltage (V_{CC}), auxiliary voltage (V_{CCX}) and bank voltage (V_{CCIO}).

There is no linear voltage regulator in devices of LV version, and V_{CCX} can be set to 1.8V, 2.5V or 3.3V. The I/O Bank voltage V_{CCIO} can be set to 1.2V, 1.5V, 1.8V, 2.5V, or 3.3V as required.

V_{CC} is 1.2V. V_{CCX} is the auxiliary power, which is used to supply some circuits in the chip, supporting 1.8V, 2.5V and 3.3V. After the chip powers on, V_{CCX} can be turned off. V_{CCIO} Bank can be set to 1.2V, 1.5V, 1.8V, 2.5V, or 3.3V as required.

Power DC Voltage Requirement

GW1NS & GW1NSR & GW1NSER series devices have several different power rails, V_{CC} , V_{CCX} and some V_{CCIO} s. In order to release the Power On Reset, V_{CC} , V_{CCX} and some special V_{CCIO} s have to be powered. Please refer to the "Power Rails Monitored by POR Circuits of Different Devices" table in Chapter 6.1 Power-up Sequence of [UG290, Gowin FPGA Products Programming and Configuration Guide](#) for required V_{CCIO} rails of different devices. V_{CCX} should be always no less than V_{CCIO} , or there will be some unexpected leakage on V_{CCIO} .

For the recommended operating range for each power voltage, refer to the "Power" sheet in the following pinouts.

- [UG824-1.3.2E GW1NS-4&4C Pinout](#)
- [UG864-1.1.1E GW1NSR-4 Pinout](#)
- [UG865-1.1.1E GW1NSR-4C Pinout](#)
- [UG883-1.1E GW1NSER-4C Pinout](#)

Power Sequence

V_{CCX} should power on no later than V_{CC} .

V_{CCX} must be greater than or equal to V_{CCIO} , otherwise V_{CCIO} will leak current.

Note!

During the power ramping procedure, if V_{CCX} is lower than V_{CCIO} , V_{CCIO} could have high current (hundreds of mA). This high current could maintain until V_{CCX} is no less than V_{CCIO} .

Recommended Reference Range of Power-up Time

Recommended reference range of power-on time for V_{CC} is 0.2ms~2ms. And the power supply ramping rate of V_{CCIO} and V_{CCX} is as shown in "Power Supply Ramp Rate" table in Chapter 3.1.3 "Power Supply Ramp Rate" of [DS821, GW1NS series of FPGA Products Datasheet](#). You

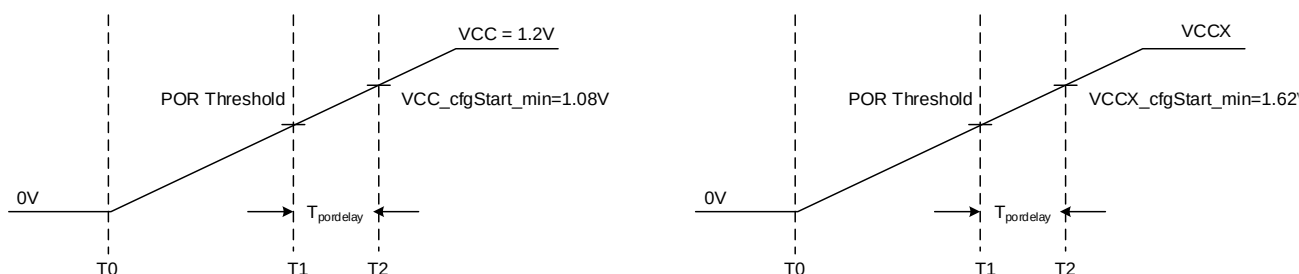
can ignore the Min. value in the “Power Supply Ramp Rate” table mentioned above (except V_{CCIO}) if you can meet the following calculation method in Power Supply Ramping Rate.

Power Supply Ramping Rate

For GW1NS & GW1NSR & GW1NSER devices, after the POR is released, the config logic will access the internal flash to read out the manufacture data for initializing. To make sure the internal flash is ready before this read operation, we have power ramping rate requirement.

The internal flash is powered by V_{CC} and V_{CCX} . As in the flash spec, $V_{CC} \geq 1.08V$ & $V_{CCX} \geq 1.62V$ is required for the reading operation. The system power rails have to meet this condition when the device starts to initializing.

Figure 1 Power Ramping Waveform



POR threshold voltage of V_{CC} and V_{CCX} is as shown in “POR Voltage” table in Chapter 3.1.5 “POR Feature” of [DS821, GW1NS series of FPGA Products Datasheet](#). POR delay time: $T_{pordelay-min}=500\mu s$, $T_{pordelay-typ}=750\mu s$, $T_{pordelay-max}=1ms$.

To calculate the ramping time requirement, first we need to know which power rail will achieve the POR trip point at last. This is T_1 , when system POR will be released. Then we can get T_2 which equals to $T_1 + T_{pordelay}$, when the device starts to access the internal flash. We need to guarantee $V_{CC} \geq 1.08V$ & $V_{CCX} \geq 1.62V$ at T_2 .

Take GW1N-4 as an example, assuming V_{CC} is powered on at last and V_{CCX} and V_{CCIO} are stable before V_{CC} , the POR release time will depend on V_{CC} rail. If the ramping time is 3.3ms, V_{CC} POR threshold is around 0.9V of GW1N-4 per “POR Voltage” table in Chapter 3.1.5 “POR Feature” of [DS100, GW1N series of FPGA Products Data Sheet](#).

$$T_1 = 3.3ms * 0.9V/1.2V = 2.475ms$$

$$T_2 = T_1 + T_{pordelay-min} = 2.975ms,$$

Thus, we can get $V_{CC_cfgstart} = 1.2V * 2.975ms/3.3ms = 1.08V$ bigger than minimum=1.08V requirement. Then it is a safe ramping rate.

Note!

The above calculation is based on the fact that the power supply is linearly lifted.

If V_{CCX} is the last rail powered on. We have to make sure it meets the $V_{CCX_cfgstart_min} = 1.62V$ requirement.

For UV devices, please use 0.3V voltage drop of the internal LDO to calculate the V_{CC} ramping time requirement, i.e. at T2, the minimum voltage should be $1.08 + 0.3 = 1.38V$.

Total Power

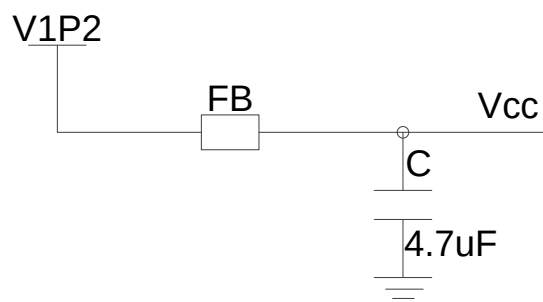
For specific density, packages, and resource utilization, GPA tools can be used to evaluate and analyze the power consumption.

Power Filter

Each FPGA power input pin is connected to the ground with a 0.1uF ceramic capacitor.

Noise processing should be noted at the input end of the V_{CC} , and the specific is as follows:

Figure 1 Noise Processing



FB is a ferrite bead, and the reference model is MH2029-221Y. The ceramic capacitance is 4.7uF, and It offers an accuracy of $\pm 20\%$.

JTAG Download

Overview

The JTAG configuration mode of Gowin FPGA products conforms to the IEEE1532 standard and the IEEE1149.1 boundary scan standard.

The JTAG configuration mode writes bitstream data to the SRAM of Gowin FPGA products. All configuration data is lost after the device is powered down. All Gowin FPGA products support the JTAG configuration mode.

Signal Description

Table 1 Signal Description

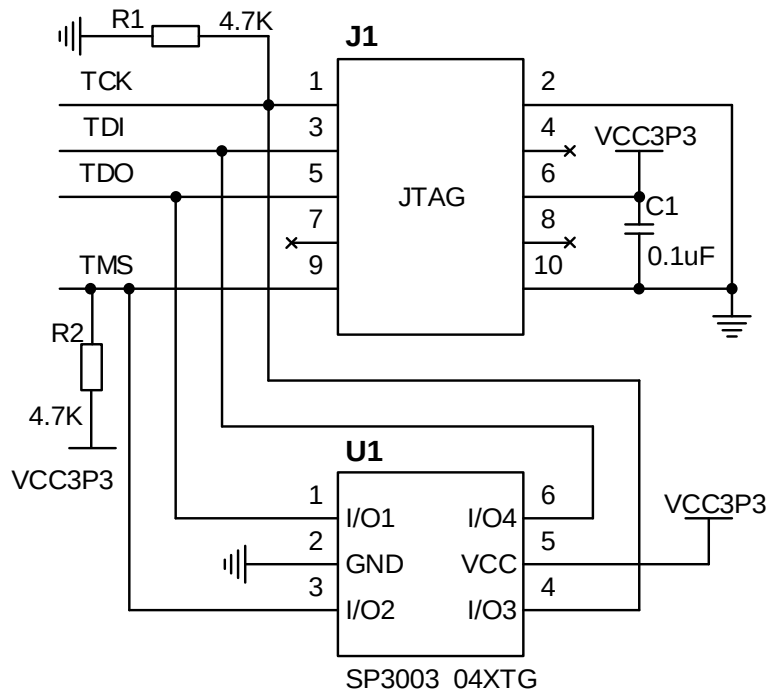
Pin Name	I/O	Description
JTAGSEL_N ^[1]	I, internal weak pull-up	Revert the JTAG pin from a GPIO to a configuration pin. Low active
TCK ^[2]	I	JTAG serial clock input
TMS	I, internal weak pull-up	JTAG serial mode input
TDI	I, internal weak pull-up	JTAG serial data input
TDO	O	JTAG serial data output

Note!

- ^[1] The JTAGSEL_N works only when the JTAG pin is set as a GPIO and the device starts to work. For the LittleBee Family of FPGA products, when MODE[2: 0]=001, the JTAGSEL_N pin is always a GPIO, in other words, the JTAGSEL_N pin and the four JTAG pins (TCK, TMS, TDI, TDO) can be used as GPIOs simultaneously; however, in this case the JTAGSEL_N pin cannot restore the JTAG pins to configuration IOs, and these pins will be restored as configuration IOs when the FPGA re-enters edit mode.
- ^[2] TCK needs to connect a 4.7K pull-down resistor on the PCB.

JTAG Circuit Reference

Figure 2 JTAG Circuit Reference



Note!

- The power of JTAG 6th pin can be set to VCC1P2, VCC1P5, VCC1P8, VCC2P5 as required.
- It is recommended to add ESD protection chip on JTAG signal for better protection of JTAG pins from electrostatic damage, optional model: SP3003_04XTG.

MSPI Download

Overview

In MSPI (Master SPI) mode, the FPGA as the Master reads bitstream data from external Flash memory via its SPI port to configure the FPGA's internal SRAM.

Signal Description

Table 2 Signal Description

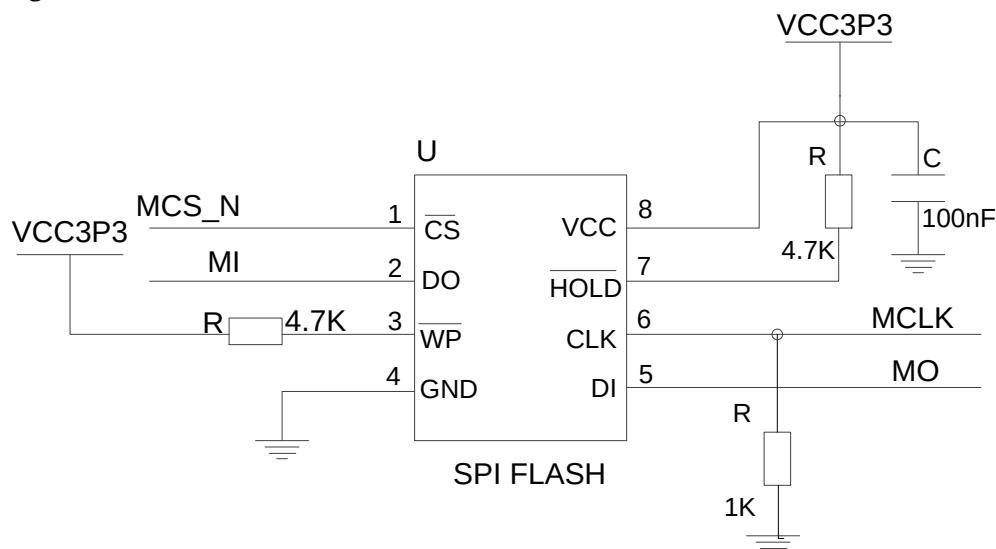
Pin Name	I/O	Description
RECONFIG_N	I, Internal weak pull-up	Low level pulse: Start Gowin CONFIG
READY	I/O	In non-JTAG configuration mode, 1'b1: The device can be programmed and configured. 1'b0: Programming configuration is prohibited
DONE	I/O	1'b1: Successfully programmed and configured. 1'b0: Programming and configuration incomplete.
MODE[2:0]	I, Internal weak pull-up	MODE select (sampled on the rising edge of READY)
MCLK	O	SPI output clock
MCS_N	O	SPI chip select, active low
MO	O	SPI output data to Slave
MI	I	SPI Input Data from Slave
FASTRD_N	I	Sampled on the rising edge of READY 1'b1: Read SPI mode (SPI instruction:0x03) 1'b0: Fast Read SPI mode (SPI instruction:0x0B)

Note!

- The MSPI configuration mode clock frequency has a tolerance of $\pm 10\%$ (Arora family) or $\pm 5\%$ (LittleBee family).
- The MSPI configuration mode clock frequency should not be greater than 66.6MHz.
- When the clock frequency is greater than 30MHz and less than 66.6MHz, you need to use the high-speed access mode of the Flash and externally pull down the FASTRD_N pin. After pulling down the FASTRD_N pin, the clock frequency needs to be greater than 5MHz.
- Leave the FASTRD_N pin floating if the clock frequency is less than 30MHz.

MSPI Circuit Reference

Figure 3 MSPI Circuit Reference



Note!

MCLK signal requires 1K pull-down resistor.

Clock Pin

Overview

The clock pins include GCLK global clock pins and PLL clock pins.

- GCLK: GCLK in FPGA products distributes in L and R quadrants. Each quadrant provides eight GCLK nets. The clock source of each GCLK can be dedicated pin or CRU, and the dedicated pin can provide better performance.
- PLL: Frequency (multiplication and division), phase, and duty cycle can be adjusted by configuring the parameters.

Signal Description

Table 3 Signal Description

Name	I/O	Description
GCLKT_[x]	I/O	Pins in global clock input, T(True), [x]: global clock No.
GCLKC_[x]	I/O	Pins for Global clock input, C(Comp), [x]: global clock No.
LPLL_T_FB/RPLL_T_FB	I	Left/Right PLL feedback input pins, T(True)
LPLL_C_FB/RPLL_C_FB	I	Left/Right PLL feedback input pins, C(Comp)
LPLL_T_IN/RPLL_T_IN	I	Left/Right PLL clock input pin, T(True)
LPLL_C_IN/RPLL_C_IN	I	Left/Right PLL clock input pin, C(Comp)

Clock Input Selection

If the external clock as PLL clock, it is recommended to input from PLL_T.

GCLK is the global clock and is connected to all resources in the device. It is recommended to input from GCLK_T.

Differential Pin

Overview

Differential transmission is a kind of signal transmission, which is different from the traditional signal line and ground line. Differential transmission signals are transmitted on these two lines. These two signals are with same amplitude, phase and opposite polarity.

Differential Type

Table 4 Differential Type

I/O Output Standard	Single/Differential	Bank V _{CCIO} (V)	Output Driver Strength (mA)
MIPI	Differential (MIPI)	1.2	N/A
LVDS25	Differential (True LVDS)	2.5/3.3	N/A
RSDS	Differential (True LVDS)	2.5/3.3	N/A
MINILVDS	Differential (True LVDS)	2.5/3.3	N/A
PPLVDS	Differential (True LVDS)	2.5/3.3	N/A
SSTL15D	Differential	1.5	8
SSTL25D_I	Differential	2.5	8
SSTL25D_II	Differential	2.5	8
SSTL33D_I	Differential	3.3	8
SSTL33D_II	Differential	3.3	8
SSTL18D_I	Differential	1.8	8
SSTL18D_II	Differential	1.8	8
HSTL18D_I	Differential	1.8	8
HSTL18D_II	Differential	1.8	8
HSTL15D_I	Differential	1.5	8
LVC MOS12D	Differential	1.2	4,8
LVC MOS15D	Differential	1.5	4,8
LVC MOS18D	Differential	1.8	4,8,12
LVC MOS25D	Differential	2.5	4,8,12,16
LVC MOS33D	Differential	3.3	4,8,12,16,24

Note!

See pinout manuals for specific differential pin positions.

RECONFIG_N, READY, DONE

Overview

RECONFIG_N, equivalent to the reset function of FPGA programming configuration, FPGA cannot perform any configuration operation when RECONFIG_N is pulled down.

As a configuration pin, a low level with a pulse width of not less than 25ns is required for GowinCONFIG configuration mode to enable the device to reload the bitstream. You can control the pin by writing logic to trigger the device to reconfigure as required.

You can configure FPGA only when the READY signal is high. The device should be restored by power on or triggering RECONFIG_N when the READY signal is low.

As an output configuration pin, it can indicate whether the FPGA can be configured currently. If the device is ready, READY signal is high. If the device fails to configure, the READY signal changes to low. As an input configuration pin, you can delay the configuration by its own logic or pulling down the READY signal.

DONE signal indicates that the FPGA is configured successfully. The signal is high after successful configuration.

As an output configuration pin, it can indicate whether FPGA configuration is successful. If configured successfully, DONE is high, and the device enters into an operating state. If the device failed to configure, the DONE signal remains low. As the input, you can delay the entry into user mode by manually pulling down the DONE signal via the logic.

When RECONFIG_N or READY signals are low, DONE signal is also low. When configuring SRAM using JTAG circuit, it does not need to take DONE signal into account.

Signal Description

Table 5 Signal Description

Name	I/O	Description
RECONFIG_N	I, internal weak pull-up	Low level pulse: start new GowinCONFIG configuration
READY ^[1]	I/O, internal weak pull-up	High-level pulse: The device can be programmed and configured
		Low-level pulse: The device cannot be programmed and configured
DONE ^[1]	I/O, internal weak pull-up	High-level pulse: Successfully programmed and configured
		Low-level pulse: Programming and configuration uncompleted or failed

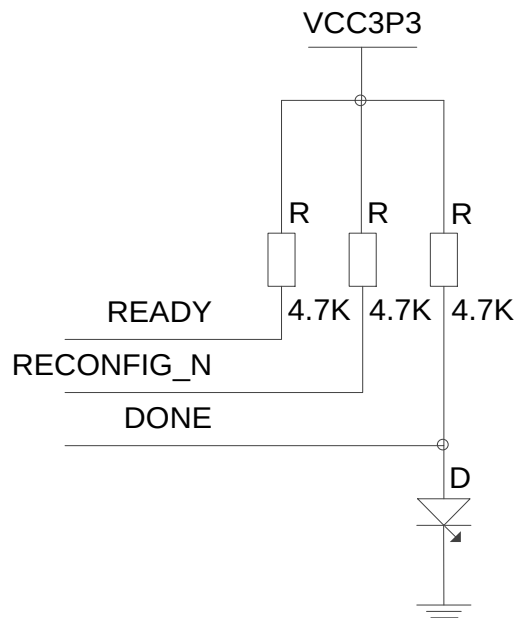
Note!

^[1] The default state of READY/DONE is open-drain output, internal weak pull-up. DONE

outputs 0 during configuration.

READY, RECONFIG_N, DONE Reference Circuit

Figure 4 READY, RECONFIG_N, DONE Reference Circuit



Note!

- The pull-up power supply is the bank voltage value VCCIO0 of the corresponding pin.
- The resistance accuracy is $\pm 5\%$.

MODE

Overview

MODE includes MODE0, MODE1, MODE2, and GowinCONFIG. When FPGA powers on or a low pulse triggers RECONFIG_N, the device enters the corresponding GowinCONFIG state according to the MODE value. MODE [2:0] is used to select the GowinCONFIG programming configuration mode. The configuration mode can be fixed by using pull-up or pull-down resistors. It is recommended to use a 4.7K resistor for pull-up or a 1K resistor for pull-down. As the number of pins for each package is different, some MODE pins are not all packaged, and the unpacked MODE pins are grounded inside. Please refer to the corresponding pinout manual for further details.

Signal Description

Table 6 Signal Description

Name	I/O	Description
MODE2	I, internal weak pull-up	GowinCONFIG mode selection pin
MODE1	I, internal weak pull-up	GowinCONFIG modes selection pin
MODE0	I, internal weak pull-up	GowinCONFIG mode selection pin

Mode Selection

Table 7 Mode Selection

Configuration Modes		MODE[2:0] ^[1]	Description
JTAG		XXX ^[2]	The LittleBee Family of FPGA products are configured via the JTAG interface by an external Host.
GowinCONFIG	AUTO BOOT	000	FPGA reads data from embedded Flash for configuration
	I ² C ^[6]	100	FPGA products are configured via the I ² C interface by an external Host.
	SSPI	001	FPGA products of the LittleBee Family are configured via the SPI interface.
	MSPI	010	As a Master, FPGA reads data from an external Flash (or other devices) via the SPI interface ^[3] .
	DUAL BOOT ^[4]	110	FPGA reads data from external Flash first and if the external Flash configuration fails, it reads from the Internal Flash.
	SERIAL ^[5]	101	External Host configures FPGA products of LittleBee Family via the DIN interface.
	CPU ^[5]	111	External Host configures FPGA products of LittleBee Family via DBUS interface.

Note!

- ^[1] The unbonded mode pins are grounded by default (except for GW1N(R)-2 and GW1N-1P5 devices, please refer to the corresponding pinout manuals).
- ^[2] The JTAG configuration mode is independent of the MODE value.
- ^[3] The SPI interfaces of the SSPI and MSPI modes are independent of each other.
- ^[4] Currently GW1N(R)-4/GW1N(R)-4B do not support DUAL BOOT.
- ^[5] The CPU configuration mode and SERIAL configuration mode share SCLK, WE_N, and CLKHOLD_N. The data bus pins for the CPU configuration mode share pins with MSPI and SSPI configuration modes.
- ^[6] Autoboot is automatically enabled in I²C mode. In I²C mode, following power-on the LittleBee devices will attempt to read data from the internal Flash first. The I²C SDA line MUST be held inactive (externally pulled-up) during Autoboot, otherwise, the device may not be configured correctly. Also, it is recommended to externally pull up the SCL line at the same time. Note that this note also applies to C version devices of which the SDA and SCL pins have internal weak pull-ups.

JTAGSEL_N

Overview

JTAGSEL_N is JTAG mode selection signal. If the JTAG pin is set to GPIO in Gowin software, the JTAG pin changes to GPIO after the device is powered on to configure successfully. If JTAG configuration fails, you can recover by pulling down JTAGSEL_N. If you do not set JTAG multiplexing, the JTAG configuration function is always available.

Signal Description

Table 8 Signal Description

Pin Name	I/O	Description
JTAGSEL_N	I, internal weak pull-up	Change JTAG pin from GPIO to configuration pin, active-low

Note!

The JTAGSEL_N pin and four JTAG pins (TCK, TMS, TDI, and TDO) configured as GPIO are exclusive:

- JTAG pins can only be used as configuration pin if JTAGSEL_N is set to GPIO.
- JTAGSEL_N can only be used as a configuration pin if JTAG pins are set to GPIOs.

FASTRD_N

Overview

It is the SPI Flash read speed selection signal in MSPI configuration mode. When FASTRD_N is high, it is normal read mode. When FASTRD_N is low, it is high speed read mode. Different manufacturers have different high-speed read operation instructions, please refer to the corresponding Flash datasheet.

Signal Description

Table 9 Signal Description

Pin Name	I/O	Description
FASTRD_N	I/O	As a configuration pin, internal weak pull-up, READY signal rising edge samples MSPI configuration speed mode; As a GPIO, it can be used as input or output.

Note!

- High level: Normal Flash mode, clock frequency should not be higher than 30MHz.
- Low level: High speed Flash mode, clock frequency should be > 30MHz and < 80MHz.

Dual-purpose Pin

Overview

To maximize the utilization of I/O, Gowin FPGA products support setting the configuration pins as GPIO pins. Before any configuration operation is performed on all series of Gowin FPGA products after power-up, all related configuration pins are used as configuration pins by default. After successful configuration, the device enters into user mode and reassigns the pin functions according to the multiplex options selected by the user.

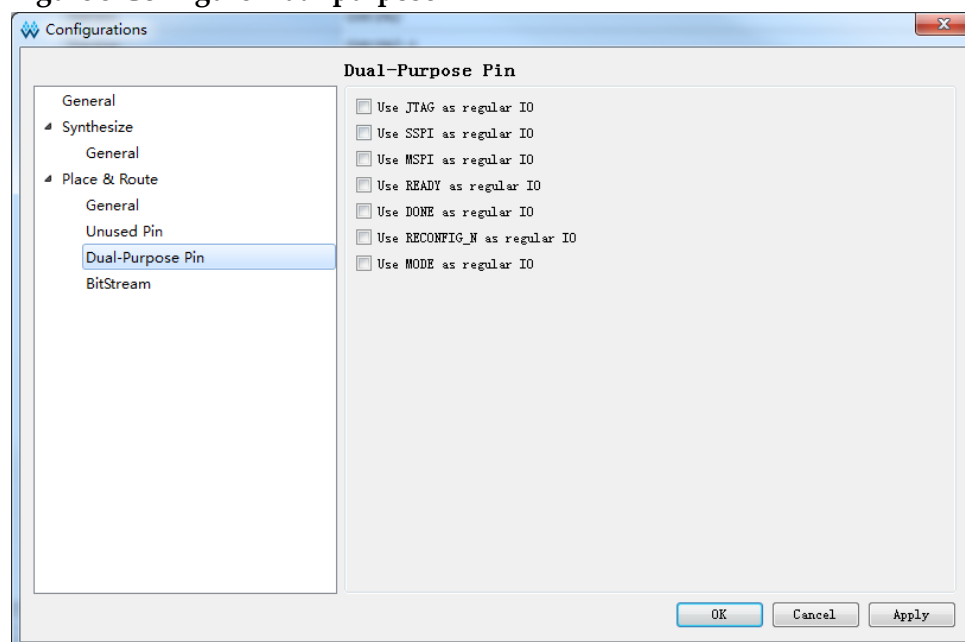
Note!

When setting the pin multiplexing option, ensure the external initial connection state of the pins does not affect the device configuration. Isolate the connections that affect the configuration first, and then wait to modify them in user mode.

The steps are as follows:

1. Open the project in Gowin Software.
2. Select "Project > Configuration > Dual-Purpose Pin" from the menu, as shown in Figure 5.
3. Check the corresponding options.

Figure 5 Configure Dual-purpose Pin



Dual-purpose Pin

- SSPI: As a GPIO, SSPI can be used as input or output.
- MSPI: As a GPIO, MSPI can be used as input or output.
- RECONFIG_N GPIO can only be used as an output. Set the initial value of RECONFIG_N as high when multiplexing it.

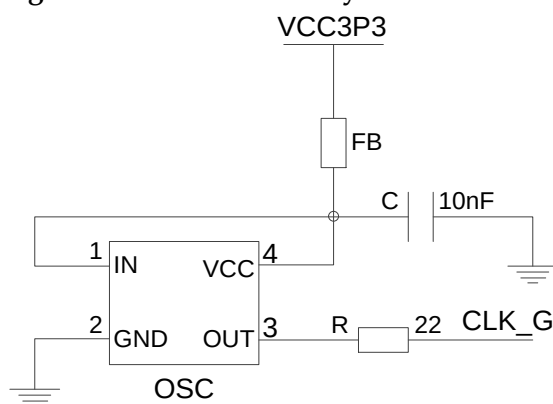
- **READY:** As a GPIO, READY can be used as input or output. As an input GPIO, the initial value of READY should be 1. Otherwise, the FPGA will fail to configure.
- As a GPIO, DONE can be used as an input or output. If DONE is used as an input GPIO, the initial value of DONE should be 1. Otherwise, the FPGA will fail to enter the user mode after configuring.
- As a GPIO, JTAG can be used as an input or output.
- As a GPIO, JTAGSEL_N can be used as an input or output.
- As a GPIO, MODE can be used as an input or output. You multiplex MODE pin, the correct value is needed to provided during configuration (power-on or low-level pulse triggers RECONFIG_N). Up to three pins can be multiplexed in MODE. Unpackaged pins are grounded internally. Please refer to pinout manuals for details. For the MODE value corresponding to different configuration modes, please refer to the corresponding device manual.

Note!

If the number of I/O ports is sufficient, use non-multiplexed pins first.

FPGA External Crystal Oscillator Circuit Reference

Figure 6 FPGA External Crystal Oscillator Circuit



FB is a ferrite bead, and MH2029-221Y is the reference model. The resistance accuracy is $\pm 5\%$, and capacitance accuracy is $\pm 10\%$.

Bank Voltage

For the Bank power supply requirements of the devices, please refer to the Power section of the following documents.

- [UG824, GW1NS-4 & 4C Pinout](#)
- [UG864, GW1NSR-4 Pinout](#)
- [UG865, GW1NSR-4C Pinout](#)
- [UG883, GW1NSER-4C Pinout](#)

Configuration Modes Supported by Each Device

GW1NS-4/4C

Table 10 GW1NS-4/4C Configuration Modes

Package	JTAG	AUTO BOOT	MSPI
CS49	Yes	Yes	No
LQ144 ^[2]	Yes	Yes	Yes
MG64	Yes	Yes	No
QN32 ^[1]	Yes	Yes	Yes
QN48	Yes	Yes	No

Note!

- ^[1] QN32 package is for GW1NS-4 devices.
- ^[2] LQ144 package is for GW1NS-4C devices.

GW1NSER-4C

Table 11 GW1NSER-4C Configuration Modes

Package	JTAG	AUTO BOOT
QN48G	Yes	Yes
QN48P	Yes	Yes

GW1NSR-4

Table 12 GW1NSR-4 Configuration Mode

Package	JTAG	AUTO BOOT
MG64P	Yes	Yes

GW1NSR-4C

Table 13 GW1NSR-4C Configuration Modes

Package	JTAG	AUTO BOOT	MSPI	DUAL BOOT
QN48P	Yes	Yes	Yes	Yes
QN48G	Yes	Yes	No	Yes
MG64P	Yes	Yes	No	No

MIPI

GW1NS series of FPGA products support embedded MIPI interface modules. BANK0/BANK1 of GW1NS-4/4C is MIPI input and BANK2 supports MIPI output.

Note!

When BANK0/BANK1 of GW1NS-4C/4 is used as MIPI input, V_{CCIO0}/V_{CCIO1} need to be set to 1.2V, and when BANK2 is used as MIPI output, V_{CCIO2} needs to be set to 1.2V; When V_{CCX} is set to 1.8 V, the speed of MIPI can only reach 60% of the speed of MIPI when V_{CCX} is set to 2.5V/3.3V.

Pin Distribution

Before the design of the circuit, you should take the FPGA pinout into consideration, and a reasonable choice should be made for IO LOGIC, global clock resource, PLL and differential signals, etc.

Note!

- During configuration, all GPIOs of the device are high-impedance with internal weak pull-ups. After the configuration is complete, the I/O state is controlled by user programs and constraints. The state of CONFIG-related I/Os varies depending on the configuration mode.
- The I/O locations of MIPI_OBUF must be assigned to differential pair pins that support the True LVDS standard.

References

- [UG290, Gowin FPGA Products Programming and Configuration User Guide](#)
- [DS821, GW1NS series of FPGA Products Datasheet](#)
- [DS861, GW1NSR Series of FPGA Products Datasheet](#)
- [DS881, GW1NSER Series of SecureFPGA Products Datasheet](#)
- [UG824, GW1NS-4&4C Pinout](#)
- [UG864, GW1NSR-4 Pinout](#)
- [UG865, GW1NSR-4C Pinout](#)
- [UG883, GW1NSER-4C Pinout](#)
- [SUG100, Gowin Software User Guide](#)

Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

Revision History

Date	Version	Description
07/28/2020	1.0E	Initial version published.
12/06/2021	1.1E	- The value described in Table 1 Recommended Range fixed. - The configuration modes supported by each device updated.
07/15/2022	1.1.1E	The note in "JTAG Circuit Reference" updated.
10/20/2022	1.2E	GW1NS-2, GW1NS-2C, GW1NSR-2, GW1NSR-2C, and GW1NSE-2C series removed.
03/10/2023	1.3E	"Power Supply" updated. "Figure 2 JTAG Circuit Reference" in "JTAG" updated. - The data of GW1N-4 example in "Power Supply Ramping Rate" updated.
03/17/2023	1.3.1E	The note in "Pin Distribution" updated.
06/29/2023	1.3.2E	"Table 10 GW1NS-4/4C Configuration Modes" in "Configuration Modes Supported by Each Device" updated. - The overview in "MODE" updated.
08/10/2023	1.3.3E	- The note in "Pin Distribution" optimized. - The mode supported by QN48 package of GW1NS-4/4C devices in "Configuration Modes Supported by Each Device" updated.
02/22/2024	1.3.4E	The note of "Table 5 Signal Description" in "RECONFIG_N, READY, DONE" added.
06/07/2024	1.4E	"References" added. "Figure 2 JTAG Circuit Reference" in "JTAG" updated. - The description of "Recommended Reference Range of Power-up Time" optimized. - The note of "FPGA External Crystal Oscillator Circuit Reference" updated.
08/09/2024	1.4.1E	The mode supported by LQ144 package of GW1NS-4/4C devices in "Configuration Modes Supported by Each Device" updated.
08/23/2024	1.4.2E	The description of "JTAG Download", "MSPI Download", and "Dual-purpose Pin" optimized.
09/26/2024	1.4.3E	The description of "Power Sequence" in "Power Supply" optimized.
11/22/2024	1.4.4E	The description of GW1NSE removed.
03/21/2025	1.4.5E	The note of MIPI_OBUF in "Pin Distribution" added.

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